

AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY



Department of Computer Science and Engineering

Course No: CSE 2104

Course Title: Data Structures Lab

Assignment No: 02

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Lab Section: A1

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Program: BSc in CSE

Question : Write a code for implementing Selection Sort.

Solution:

Selection Sort: An in-place, comparison-based sorting algorithm that repeatedly finds the minimum element from an unsorted portion of a list and swaps it with the first element of that portion, running in $O(n^2)$ time.

Implementation of Selection Sort :

/*-array of books with its title,publication year, author's name

-year compare korle descending order,same year publication hole author name lexicography(basically alphabet order with numbers in ascending order)

-selectiong sorting use korte hobe in ascending order

*/

#include <bits/stdc++.h>

using *namespace* std;

struct bookInfo

{

int publicationYear;

string title;

string author;

};

void selectionSort(bookInfo books[], *int* n)

{

 for (*int* i = 0; i < n - 1; i++)

 {

int maxIndex = i;

 for (*int* j = i + 1; j < n; j++)

```

{
    if (books[j].publicationYear > books[maxIndex].publicationYear)
    {
        maxIndex = j;
    }
    else if (books[j].publicationYear == books[maxIndex].publicationYear)
    {
        if (books[j].author < books[maxIndex].author)
        {
            maxIndex = j;
        }
        else if (books[j].author == books[maxIndex].author)
        {
            if (books[j].title < books[maxIndex].title)
            {
                maxIndex = j;
            }
        }
    }
}
swap(books[i], books[maxIndex]);
}

```

```

cout << "Sorted book information: " << endl;

```

```

for (int i = 0; i < n; i++)

```

```

    {
        cout << books[i].title << ", "
        << books[i].author << ", "
        << books[i].publicationYear << endl;
    }
}

int main(void)
{
    int n;
    cout << "Please enter the number of books: " << endl;
    cin >> n;
    bookInfo books[n];
    if (1 <= n && n <= 100)
    {
        cout << "Please enter the book information: " << endl;
    }
    else
    {
        cout << "Invalid input" << endl;
        return 0;
    }
    for (int i = 0; i < n; i++)
    {
        cout << "Please enter the book title: " << endl;
        getline(cin>>ws, books[i].title);
    }
}

```

```
if (1 > books[i].title.length() || books[i].title.length() > 100)
{
    cout << "Invalid input" << endl;
    return 0;
}
cout << "Please enter the book's author: " << endl;
getline(cin>>ws, books[i].author);
if (1 > books[i].author.length() || books[i].author.length() > 100)
{
    cout << "Invalid input" << endl;
    return 0;
}
cout << "Please enter the book publication year: " << endl;
cin >> books[i].publicationYear;
if (1900 > books[i].publicationYear || books[i].publicationYear > 2025)
{
    cout << "Invalid input" << endl;
    return 0;
}
}
selectionSort(books, n);
}
```

Input Samples :

>>> Example 1

Please enter the number of books:

1111

Invalid input

>> Example 2

Please enter the number of books:

3

Please enter the book information:

Please enter the book title:

Last of Us

Please enter the book's author:

Neil Druckaman

Please enter the book publication year:

2013

Please enter the book title:

Resident Evil 2 Remake

Please enter the book's author:

Hideaki Itsuno

Please enter the book publication year:

2019

Please enter the book title:

Devil May Cry 5

Please enter the book's author:

Hideaki Kamiya

Please enter the book publication year:

2019

Sorted book information:

Resident Evil 2 Remake, Hideaki Itsuno, 2019

Devil May Cry 5, Hideaki Kamiya, 2019

Last of Us, Neil Druckaman, 2013